

PORTLAND BUOY 44007, ME, USA

WMO: 992780

Lat: 43.525N

Lon: 70.141W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-10.6	-20.2	0.6	-12.2	-18.2	0.8	-9.4	16.8	-1.3	15.6	-2.2	7.9	310	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	22.2	20.0	21.2	19.3	20.4	18.8	21.0	21.5	20.1	20.7	19.4	20.0	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.4	21.1	19.9	14.6	20.4	19.1	13.9	19.7	60.8	21.4	57.6	20.5	55.1	20.1	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	13.1	11.1	DB	-16.1	25.2	2.7	2.0	-18.1	26.6	-19.6	27.8	-21.1	28.9	-23.1	30.3	(4)
(5)			WB	-16.6	21.5	2.7	1.6	-18.5	22.6	-20.1	23.5	-21.6	24.4	-23.6	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-1.9	-0.7	1.8	6.0	10.2	14.7	17.9	18.1	15.6	11.1	6.3	1.8	(6)
(7)		DBStd	7.57	4.98	4.04	3.47	2.49	2.40	2.34	1.80	1.54	2.10	2.69	3.41	4.17	(7)
(8)		HDD10.0	1463	369	299	256	122	27	0	0	0	0	20	116	253	(8)
(9)		HDD18.3	3643	627	533	513	371	253	111	31	23	84	225	361	512	(9)
(10)		CDD10.0	898	0	0	1	2	33	141	244	251	168	53	6	1	(10)
(11)		CDD18.3	36	0	0	0	0	0	2	16	16	3	0	0	0	(11)
(12)		CDH23.3	11	0	0	0	0	0	0	7	3	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.5	7.3	6.7	6.5	5.5	4.6	4.1	3.7	3.8	4.6	6.1	6.5	7.0	(14)
(15)	Precipitation	PrecAvg	1248	84	87	107	106	96	115	94	89	107	134	113	116	(15)
(16)		PrecMax	1744	161	207	259	243	266	276	208	307	274	393	179	183	(16)
(17)		PrecMin	854	7	41	26	5	24	28	39	12	30	25	34	43	(17)
(18)		PrecStd	198	33	35	52	55	53	62	41	59	60	83	37	38	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	7.6	11.4	14.4	18.7	22.0	24.2	23.5	21.9	18.5	13.9	11.6	(19)
(20)			MCWB	9.0	6.2	8.8	10.4	14.0	17.0	20.4	21.7	19.5	15.5	12.5	9.9	(20)
(21)		2%	DB	8.0	6.4	8.3	11.2	16.1	19.9	22.4	22.0	20.2	16.5	12.7	9.9	(21)
(22)			MCWB	6.7	5.3	6.4	9.1	13.4	16.7	19.9	20.3	18.4	14.8	11.0	8.6	(22)
(23)		5%	DB	6.5	5.3	6.9	10.0	14.7	18.7	21.3	21.1	19.2	15.4	11.8	8.9	(23)
(24)			MCWB	5.3	4.1	5.3	8.0	12.3	16.5	19.6	19.4	17.6	13.9	10.4	7.6	(24)
(25)		10%	DB	4.9	4.4	5.7	9.0	13.4	17.8	20.4	20.4	18.5	14.6	11.0	7.6	(25)
(26)			MCWB	3.3	3.0	4.3	7.4	11.4	16.2	19.2	18.9	17.1	13.2	9.7	6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	7.1	10.1	11.9	16.2	19.2	22.0	22.9	21.5	16.8	13.5	10.6	(27)
(28)			MCDWB	9.5	7.4	10.2	13.0	16.8	20.0	22.7	23.4	21.8	17.4	13.9	11.1	(28)
(29)		2%	WB	7.2	5.6	7.1	10.2	14.2	18.0	20.7	21.4	19.3	15.3	11.9	9.1	(29)
(30)			MCDWB	7.7	5.9	7.8	10.8	15.4	18.7	21.2	21.8	19.9	15.7	12.4	9.7	(30)
(31)		5%	WB	5.3	4.5	5.8	8.9	12.8	17.3	20.0	20.3	18.5	14.5	11.0	8.0	(31)
(32)			MCDWB	6.3	5.1	6.6	9.6	13.7	18.1	20.6	20.8	19.1	15.0	11.5	8.7	(32)
(33)		10%	WB	3.5	3.4	4.8	8.0	11.7	16.6	19.3	19.6	17.6	13.6	10.1	6.6	(33)
(34)			MCDWB	4.9	4.3	5.7	8.9	12.6	17.5	20.0	20.1	18.3	14.2	10.9	7.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	5.4	4.6	3.7	3.5	3.4	3.5	3.4	3.3	3.8	4.2	5.3	(35)
(36)			MCDBR	6.3	5.2	5.1	5.2	5.9	5.4	5.0	4.4	3.9	4.1	4.2	6.0	(36)
(37)			MCWBR	6.6	5.5	4.4	4.1	4.2	3.3	3.2	3.5	3.3	3.8	4.6	5.7	(37)
(38)		5% WB	MCDWR	6.1	5.0	4.8	4.8	5.0	4.5	4.1	4.2	3.6	3.5	3.9	5.6	(38)
(39)			MCWBR	6.7	5.7	4.7	4.4	4.1	3.4	3.1	3.6	3.4	3.9	4.3	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.308	0.331	0.358	0.392	0.444	0.457	0.425	0.380	0.341	0.321	0.299	(40)	
(41)		taud	2.473	2.458	2.452	2.420	2.370	2.253	2.230	2.308	2.410	2.512	2.531	2.528	(41)	
(42)		Ebn at Noon	839	892	911	908	881	833	816	831	846	842	806	806	(42)	
(43)		Edh at Noon	72	86	98	110	119	135	136	122	102	81	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.66	2.55	3.68	4.65	5.21	5.58	5.89	5.35	4.29	2.78	1.80	1.34	(44)	
(45)		RadStd	0.11	0.19	0.23	0.49	0.58	0.46	0.38	0.27	0.24	0.25	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79	N/A

Nomenclature: See separate page